Sparrta-Small

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1 Hypotheses how results should look

H1: Front/back relation should give better (maybe ideal) results. Different sizes and occlusion relations intuitivelly should be easier for model to recognise than left/right ones.

H2: Left/right relation should be on the level of chance at first layer of ResNet and (little bit better) on ViT. First layer(-s) should produce localised features (about the objects themself) rather than relation (between objects) features. In ResNet its an architecture (far patches phisically are not connected yet), in ViT not, but intuitivelly should be true.

H3: CLS token should produce worse results than average pooled token. Because CLS token is deliberately made for classification (or image descibing in case of DINO), not spatial (geometry) understanding. The classification goal of pretraining could possible also led to worse spatial understanding at later layers. So I expect peak somewhere in the middle layers.

H4: DINO-ViT should give better results than Supervised-ViT. Supervised ViT is pretrained strictly for classification, whether DINO is learning relations through its training architectures. Its shown that last layers of DINO attention maps are good at segmentation, which is, presumably, gives better spatial understanding.

2 Setup

Dataset: 250 images for each relation. 4 training/validation colors for figures, 2 different colors for test split. Background color is random bright. Size/position jittering as asked in statement.

Feature extraction: Official preprocessing(with its own mean and std, but without random crop) is applied for every model. For ViT-supervised and ViT-DINO we extract features after each of the 12 blocks (residual streams). Since those blocks are the same, transformations are also. Layer 12 is extracted before head and its LayerNorm, so it is the same as other blocks. CLS is token 0, for pooling we use average pooling across 196 patch tokens. ResNet features are taken after each 'stage', and than followed by global average pooling. The same procedure for every stage.

Normalization before probing: Between feature extraction and probing, StandardScaler is calculated among each feature on training dataset. It is applied for val and test sets, so its the same across one seed and does not intersect with val or test data.

Probing: Train and validation sets are distinguished based on 3 seeds(70/10/20), head is trained using sklearn LogisticRegression on training dataset with 11 different C values. The C with highest validation accuracy than chosen and probe is evaluated on test set. For binary probes we use the same logic but with 500 chosen images.

3 Results

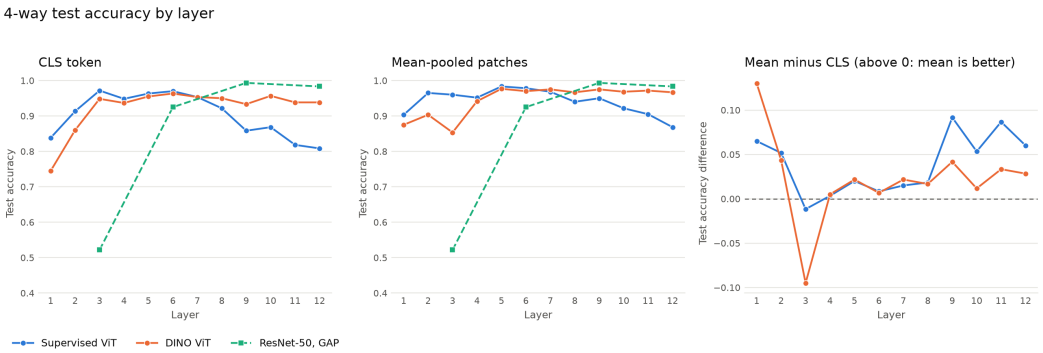


Figure 1: Results for 4-way relation.

Supervised ViT has better results on average than DINO, but peak accuracy is almost the same comparing DINO and Supervised ViTs. ResNet is better than two ViTs models. **Hypothesis 4 is rejected** for 4way test.

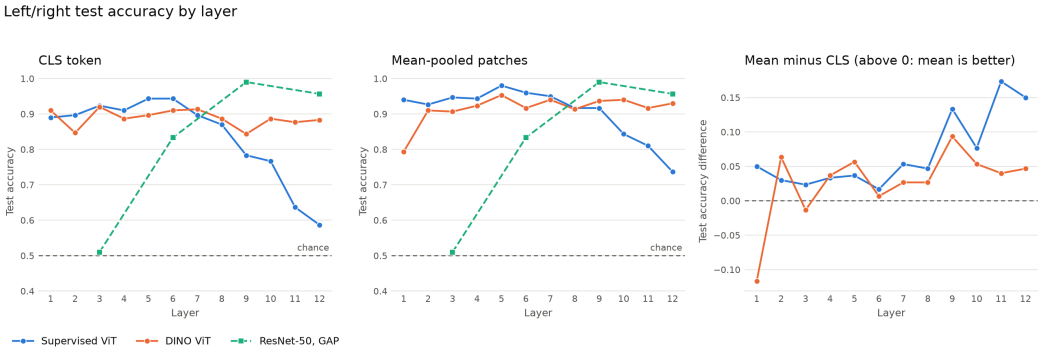


Figure 2: Results for left-right relation.

ResNet starts on chance level, which **supports half of Hypothesis 2**. Other half(for ViTs) is rejected, but the reason is presumably dataset and not bad reasoning(see Limitation section). Mean pooling is still better than CLS, so **Hypothesis 3 is supported**. A sharp decline of accuracy at the end of SUP-ViT layers is conserning. My best explanation is the classification objective of ViT pretraining. So that relations are fading away with new layers. This explains why CLS decrease is much bigger than mean-pooled. And it also supported by the fact that DINO pretraining objective contains more spatial understanding (it need to match different crops), thats why DINO decrease is lower.

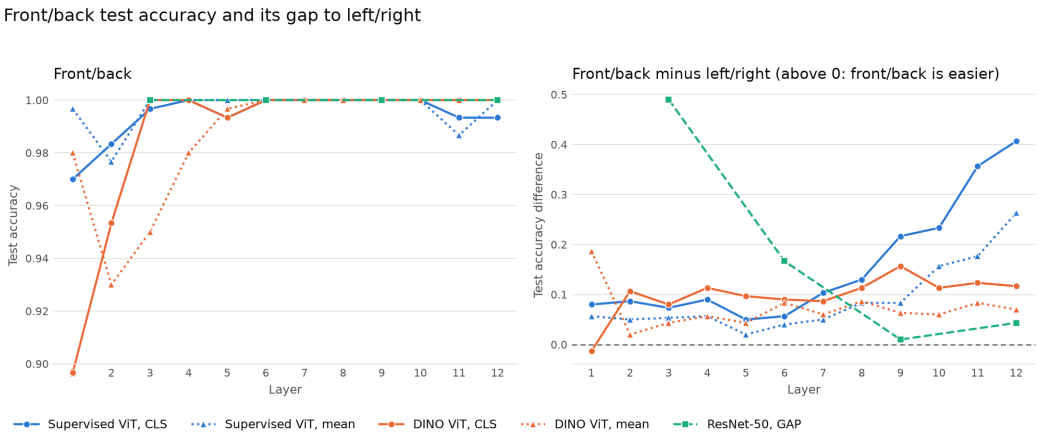


Figure 3: Results for front-back relation.

This clearly **proves Hypothesis 1**. Probably no real geometry is ever involved in front/back relation understanding.

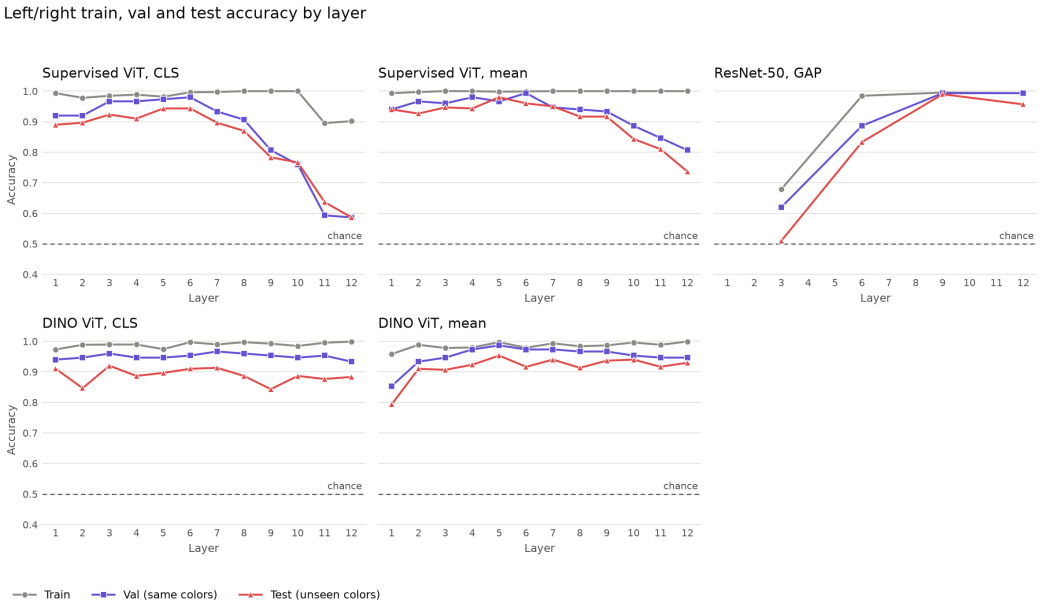


Figure 4: Comparison of train, validation, and test splits.

This is the most conserning graphics. They may prove that this dataset is not really suitable for this kind of task. More about it in limitations section.

Table 1: Peak values across different relations.

model	task	peak layer	**test acc**	std
Supervised ViT, CLS	4-way	3	**0.9717**	0.0076
Supervised ViT, CLS	Left/right	5	**0.9433**	0.0115
Supervised ViT, CLS	Front/back	4	**1.0000**	0.0000
Supervised ViT, mean	4-way	5	**0.9833**	0.0076
Supervised ViT, mean	Left/right	5	**0.9800**	0.0100
Supervised ViT, mean	Front/back	3	**1.0000**	0.0000
DINO ViT, CLS	4-way	6	**0.9633**	0.0161
DINO ViT, CLS	Left/right	3	**0.9200**	0.0346
DINO ViT, CLS	Front/back	3	**1.0000**	0.0000
DINO ViT, mean	4-way	5	**0.9767**	0.0076
DINO ViT, mean	Left/right	5	**0.9533**	0.0153
DINO ViT, mean	Front/back	6	**1.0000**	0.0000
ResNet-50, GAP	4-way	3	**0.9933**	0.0029
ResNet-50, GAP	Left/right	3	**0.9900**	0.0100
ResNet-50, GAP	Front/back	1	**1.0000**	0.0000

4 Limitations

Figure 4 shows that training, validation and test accuracies are almost perfect at the first layers. Test accuracy shows that dataset is too small and one linear layer is enough to give perfect match(only 700 images for 4 classes with 768 features is enough to fit it despite actual quality of features), but than val/test accuracies at first layer(where model presumably did not learn spatial understanding yet) shows that this understanding is not necessary. I assume that positional embedding is already enough for probe to classify position relation by itself. Due to the fact that reference object is in the center (with small jitter), position of target object can tell relation by itself. Example: target object center with x=150 is telling that the relation is 'left' without any spatial understanding. So we need to increase number of pictures in dataset, and change dataset generation logic so that global position will not give away relation by itself.